

(a) Calculate the initial acceleration of the motorcycle.

From the graph, the motorcycle reaches a velocity of 20.0 m s^{-1} in 4.0 s , starting from rest.

$$a = \frac{v - u}{t} = \frac{20.0 - 0}{4.0} = 5.0\text{ m s}^{-2}$$

(b) Determine the distance between the car and motorcycle at 4.0 s .

- Car is moving at constant speed:

$$s = vt = 15.0 \times 4.0 = 60.0\text{ m}$$

- Motorcycle is accelerating from rest:

$$s = \frac{1}{2}at^2 = \frac{1}{2} \times 5.0 \times 4.0^2 = 40.0\text{ m}$$

$$\text{Distance between them} = 60.0 - 40.0 = 20.0\text{ m}$$

(c)(i) Determine the frictional force acting on the motorcycle at $t = 2.0\text{ s}$

Given:

- Acceleration $a = 5.0\text{ m s}^{-2}$
- Mass $m = 290\text{ kg}$
- Engine force $F_{\text{engine}} = 1800\text{ N}$

$$F_{\text{net}} = ma = 290 \times 5.0 = 1450\text{ N}$$

$$f = F_{\text{engine}} - F_{\text{net}} = 1800 - 1450 = 350\text{ N}$$

(c)(ii) Explain why the driving force must be increased with time to maintain constant acceleration.

As the motorcycle speeds up, **resistive forces** like **air resistance** increase.
To maintain constant acceleration (i.e. constant **net** force), the **engine force must increase** to compensate.

(d) Extend the velocity-time graph from 4.0 s to 10.0 s .

From the question:

- Velocity at $4.0\text{ s} = 20.0\text{ m/s}$
- Between 4.0 s and 5.0 s , the driving force increases, causing a further acceleration
- After 5.0 s , the driving force becomes constant $\Rightarrow$ no net force $\Rightarrow$ constant velocity

So:

- Draw a straight line from $(4.0\text{ s}, 20.0\text{ m/s})$ to $(5.0\text{ s}, 25.0\text{ m/s})$
- Then a horizontal line from $(5.0\text{ s to } 10.0\text{ s})$ at 25.0 m/s

 Revision Tips

- Use $a = \frac{v-u}{t}$ when acceleration is constant
- To find distance from a **v–t graph**, calculate **area under the graph**
- Total force = **engine force minus friction**
- To keep accelerating as resistive forces grow, **engine force must grow too**
- After net force becomes zero, motion continues at **constant velocity**